Experiment 2

Preparation of a Software Project Plan Covering Scope, Timeline, and Team Roles for a Smart Parking Management System

Aim

To prepare a comprehensive software project plan for the development of a Smart Parking Management System, including project scope, schedule, resources, team responsibilities, risk analysis, and project deliverables.

Objectives

- Understand software project planning.
- Define project scope.
- Estimate project schedule.
- Allocate team responsibilities.
- Identify project risks.
- Prepare a complete project plan

Introduction

Urban vehicle growth makes finding parking difficult, causing congestion and waste. Traditional manual systems are inefficient and error-prone.

The **Smart Parking Management System** automates operations, allowing users to find, reserve, and pay for slots while providing administrators with real-time monitoring and reporting tools.

By leveraging cloud and mobile technologies, this project enhances parking efficiency, user convenience, and reduces operational costs.

Project Overview

- **Core Purpose:** The system is designed to simplify finding and managing parking spaces, ultimately reducing traffic congestion caused by drivers searching for parking.
- **Key Functionality:** Users can reserve parking slots in advance before reaching their destination.
- **System Modules:** The software includes modules for user management, parking slot management, online payments, notifications, and administrative controls.
- **Real-Time Monitoring:** Users have instant visibility into available parking spaces.
- **Security:** The system uses QR-code-based verification to ensure secure entry and exit from parking facilities.
- **Accessibility:** The software is designed to be accessible via both mobile and web applications for user flexibility

Project Scope

In Scope

The project scope defines the core features included in the system to ensure a seamless parking experience. The Smart Parking Management System will provide the following functionalities:

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- User Registration and Authentication
- Secure Login
- Search Available Parking Slots
- Online Slot Reservation
- QR Code Generation
- Digital Payment Integration
- Parking History
- Email and SMS Notifications

- ❖ Administrator Dashboard
- ❖ Parking Reports
- ❖ User Profile Management
- ❖ News and Announcements Feed

These features are essential for delivering an efficient, user-centric parking management solution.

Out of Scope

To ensure project focus and timely delivery, the following non-core features are excluded from this project scope:

- ❖ Vehicle servicing
- ❖ Traffic signal management
- ❖ Toll collection systems
- ❖ Vehicle insurance management
- ❖ Automatic driving assistance

Clearly defining these boundaries helps avoid unnecessary feature creep and keeps the development efforts aligned with the primary objectives.

Functional Requirements

Functional requirements describe the operations that the Smart Parking Management System must perform. The application should allow users to create accounts, log in securely, search for available parking slots, reserve parking spaces, and make online payments. It should also send booking confirmations and notifications to users. The administrator should be able to manage parking slots, user information, booking records, and payment details. These requirements define the core functionality of the application and guide the development process.

Module	**Functions**

User	Registration, Login, Slot Booking
Parking	Slot Allocation, Availability
Payment	Online Payment, Receipt
Admin	Manage Users and Parking

Non-Functional Requirements

Non-functional requirements define the quality and performance of the Smart Parking Management System. The application should provide fast response times and maintain high system availability. It should ensure secure user authentication and protect payment information using encryption. The system must support multiple users simultaneously without affecting performance. It should also be easy to maintain, update, and expand as new features are introduced. These requirements improve the reliability, security, and overall usability of the software.

Requirement	**Description**
Performance	Response within 2 seconds
Security	Encrypted Login & Payments
Reliability	99% System Availability
Scalability	Supports Multiple Users
Maintainability	Easy Updates and Maintenance

Team Roles and Responsibilities

A successful software project requires collaboration among team members with different responsibilities. The **Project Manager** oversees planning, scheduling, and overall project coordination. The **Business Analyst** gathers and documents user requirements. The **Developers** are responsible for coding and implementing application features, while the **Database Administrator** manages the database. The **Tester** verifies software quality by identifying and reporting defects. The **DevOps Engineer** handles deployment and system maintenance. Clearly assigning roles improves teamwork and ensures smooth project execution.

Role	Responsibility
Project Manager	Project Planning
Developer	Coding
Tester	Software Testing
DBA	Database Management

Project Timeline

The project is planned to be completed in **8 weeks**. The work starts with planning and requirement analysis, followed by design, development, testing, deployment, and maintenance. A proper schedule helps the team complete tasks on time.

Work Breakdown Structure (WBS)

The Work Breakdown Structure divides the project into smaller tasks such as planning, requirement analysis, design, development, testing, deployment, and maintenance. It helps the team organize work, monitor progress, and complete the project efficiently.